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AWS Certified Database Specialty (DBS-C01)

Interview Preparation Short Notes

7 Chapters • 37 Topics • Cloud DBA Reference

Design | Migration | Operations | Monitoring | Security

Note: DBS-C01 was retired 30 Apr 2024 — this is an interview / knowledge reference

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Table of Contents

Table of Contents.....	2
Chapter 1 — Workload-Specific Database Design (Choosing the Right DB).....	4
1.1 Database Selection Framework.....	4
1.2 Amazon RDS Deep Dive	4
1.3 Amazon Aurora Deep Dive	4
1.4 Amazon DynamoDB Deep Dive	5
1.5 ElastiCache & MemoryDB.....	5
1.6 Purpose-Built Databases	5
1.7 Redshift & Analytics Data Stores.....	6
Chapter 2 — High Availability, Scalability & Disaster Recovery.....	7
2.1 RDS Multi-AZ vs Read Replicas	7
2.2 Aurora HA, Global Database & Failover	7
2.3 DynamoDB Global Tables & Scaling	7
2.4 Caching & Read-Scaling Strategies	7
2.5 Backup, Restore & Point-in-Time Recovery	8
2.6 Disaster Recovery Strategies for Databases.....	8
Chapter 3 — Deployment & Migration.....	9
3.1 Deployment Automation.....	9
3.2 Migration Strategy: Homogeneous vs Heterogeneous.....	9
3.3 AWS DMS Deep Dive	9
3.4 AWS Schema Conversion Tool (SCT)	9
3.5 Migration Patterns & Minimizing Downtime	10
Chapter 4 — Management & Operations.....	11
4.1 Maintenance Windows & Engine Upgrades.....	11
4.2 Parameter & Option Groups, Configuration	11
4.3 Storage Management & Scaling	11
4.4 RDS Proxy & Connection Management.....	11
4.5 Operational Automation	12
Chapter 5 — Monitoring & Troubleshooting.....	13
5.1 CloudWatch Metrics for Databases.....	13
5.2 Performance Insights & Enhanced Monitoring	13
5.3 RDS / Aurora Troubleshooting	13
5.4 DynamoDB Troubleshooting	13
5.5 Logging & Auditing	14
Chapter 6 — Database Security	15

6.1 Encryption at Rest & In Transit.....	15
6.2 IAM Authentication & Access Control.....	15
6.3 Network Security	15
6.4 Secrets Management & Rotation	15
6.5 Auditing, Compliance & Governance	16
Chapter 7 — Certification Status, Reference & Interview Prep.....	17
7.1 DBS-C01 Status & Domain Review	17
7.2 Service Selection Decision Guide	17
7.3 Key Limits & Numbers Reference.....	17
7.4 Interview Strategy & Career Path.....	17

Chapter 1 — Workload-Specific Database Design (Choosing the Right DB)

1.1 Database Selection Framework

- Match the data model + access pattern to the engine, not the other way around. Relational (ACID, joins, complex queries) → RDS/Aurora. Key-value at scale → DynamoDB. In-memory → ElastiCache/MemoryDB.
- Purpose-built: document → DocumentDB, wide-column → Keyspaces (Cassandra), graph → Neptune, time-series → Timestream, ledger/immutable → QLDB, search → OpenSearch, warehouse/OLAP → Redshift.
- Key questions: OLTP vs OLAP? Structured/semi/unstructured? Query patterns known & fixed (DynamoDB) or ad-hoc (relational)? Scale (GB vs PB)? Consistency needs? Latency target?

TIP Interview default: 'known access patterns + massive scale + single-digit ms' = DynamoDB; 'complex joins/reporting' = Aurora/RDS; 'analytics at scale' = Redshift.

1.2 Amazon RDS Deep Dive

- Managed engines: MySQL, PostgreSQL, MariaDB, Oracle, SQL Server (+ Aurora). AWS manages patching, backups, Multi-AZ, monitoring — no OS/SSH access (except RDS Custom for Oracle/SQL Server).
- Instance classes: db.t (burstable), db.m (general), db.r/db.x (memory-optimized). Storage: gp3 (default), io1/io2 (provisioned IOPS), magnetic (legacy). Storage autoscaling avoids ENOSPC.
- Parameter groups = engine config (e.g. max_connections); Option groups = extra features (Oracle/SQL Server TDE, SQLSERVER_AUDIT, native backup/restore). Static params need a reboot.
- RDS Custom gives OS + database access for apps needing custom configs/agents (Oracle & SQL Server). Blue/Green Deployments stage + switch over for near-zero-downtime upgrades.

WATCH SQL Server on RDS: no in-place engine-edition downgrade, limited to supported features; use native backup/restore to S3 via option group for large DB moves.

1.3 Amazon Aurora Deep Dive

- MySQL/PostgreSQL-compatible, distributed storage: 6 copies across 3 AZs, 4/6 write quorum + 3/6 read quorum, self-healing, storage auto-grows in 10 GB increments to 128 TiB.
- Up to 15 low-lag reader replicas share the storage volume (replica lag typically <10 ms — no binlog shipping). Endpoints: cluster (writer), reader (load-balanced readers), custom (subset routing), instance.
- Failover ~30 s (promotes a reader by priority tier). Aurora Serverless v2 scales in fine ACU increments for variable load. Aurora Global Database = cross-region <1 s replication.
- Aurora features: Backtrack (MySQL — rewind cluster in-place, no restore), fast clones (copy-on-write), parallel query, Babelfish (run SQL Server T-SQL/TDS on Aurora PostgreSQL).

TIP Aurora vs RDS: choose Aurora for higher throughput, faster/cleaner failover, 15 readers, and cloud-native DR; RDS for Oracle/SQL Server or lower baseline cost.

1.4 Amazon DynamoDB Deep Dive

- Serverless NoSQL, single-digit ms, multi-AZ by default. Primary key = partition key (hash) or partition + sort (composite). Item max 400 KB. Auto-partitioned by partition-key hash.
- Capacity: On-Demand (spiky/unknown, pay-per-request) vs Provisioned (RCU/WCU + auto-scaling, cheaper for steady). 1 RCU = 1 strong 4 KB read/s (2 eventually consistent); 1 WCU = 1x1 KB write/s.
- Indexes: GSI (any keys, own capacity, eventually consistent, add anytime) vs LSI (same PK + alt sort key, strong consistency, must create at table creation, 10 GB item-collection cap).
- Design single-table with high-cardinality partition keys to avoid hot partitions; use key overloading + GSI overloading for multiple access patterns. Adaptive capacity mitigates skew.

WATCH Avoid Scan on large tables (full read); model for Query. Hot partition keys → throttling even under total provisioned capacity.

1.5 ElastiCache & MemoryDB

- ElastiCache Redis (or Valkey): replication, Multi-AZ auto-failover, cluster-mode sharding, persistence (RDB/AOF), pub/sub, sorted sets — for HA caching, sessions, leaderboards, rate limiting.
- ElastiCache Memcached: simple, multi-threaded, horizontally scalable, no persistence/replication — pure caching where data loss is acceptable.
- MemoryDB for Redis = durable in-memory PRIMARY database (multi-AZ transaction log, 11 9s), not just a cache. Use when you need Redis speed as the system of record.
- Caching patterns: lazy-loading (cache on miss, stale risk) vs write-through (write cache+DB, always fresh, write overhead). TTLs bound staleness.

TIP Cache-aside (lazy) is the common default; add write-through for read-heavy data that must be fresh. Redis for HA + rich types; Memcached for simple scale.

1.6 Purpose-Built Databases

- DocumentDB = managed MongoDB-compatible document store (separates storage/compute like Aurora). Keyspaces = serverless Cassandra (CQL). Neptune = graph (Gremlin/openCypher/SPARQL) for fraud/social/recommendations.
- Timestream = serverless time-series (IoT/metrics) with memory + magnetic tiers. QLDB = immutable, cryptographically verifiable ledger (append-only journal).
- Match to workload: relationships/traversal → Neptune; time-ordered metrics → Timestream; verifiable history/audit → QLDB; flexible JSON docs → DocumentDB; Cassandra migration → Keyspaces.

TIP Don't force relational for graph/time-series/ledger — purpose-built engines give better performance + lower ops for those access patterns.

1.7 Redshift & Analytics Data Stores

- Redshift = MPP columnar OLAP warehouse (PostgreSQL-based). RA3 nodes decouple compute from managed storage; Serverless auto-provisions. NOT for OLTP.
- Tuning: distribution style (KEY co-locates join rows, ALL replicates small dims, EVEN default), sort keys (zone maps skip blocks), WLM/queues, result cache, materialized views.
- Redshift Spectrum queries S3 without loading; data sharing across clusters/accounts (no copy); zero-ETL from Aurora; COPY for bulk load from S3.

TIP Athena = serverless ad-hoc SQL on S3 (pay-per-scan); Redshift = repeated complex analytics at scale. Both complement a data lake (S3 + Glue Catalog).

Chapter 2 — High Availability, Scalability & Disaster Recovery

2.1 RDS Multi-AZ vs Read Replicas

- Multi-AZ (HA): synchronous standby in another AZ, automatic failover via DNS CNAME flip (~60–120 s). Standby is NOT readable (classic Multi-AZ). For availability, not scaling.
- Multi-AZ DB cluster (newer): 1 writer + 2 readable standbys across 3 AZs, semi-sync commit, faster failover (~35 s) and readable standbys — more capacity + resilience.
- Read Replicas (scaling): asynchronous, up to 15 per source (engine-dependent), can be cross-region, can be promoted to standalone. Use for read scaling, reporting, and cross-region DR.

TIP Multi-AZ = availability (automatic failover). Read Replica = read scalability (+ manual DR promotion). Combine both for read-heavy HA.

2.2 Aurora HA, Global Database & Failover

- Aurora replicas double as failover targets (assign failover priority tiers 0–15; lowest tier + largest size wins). Failover ~30 s; reader endpoint auto-balances read traffic.
- Aurora Global Database: one primary region + up to 5 secondaries, storage-level replication typically <1 s, RPO ~1 s, RTO <1 min on managed planned failover; write forwarding for secondary writes.
- Storage is self-healing and fault-tolerant (survives AZ loss with quorum). Backtrack (MySQL) rewinds the cluster in place to recover from logical errors without a restore.

TIP Cross-region DR for relational at scale → Aurora Global Database (fast, low RPO/RTO). Single-region HA is built into the Aurora storage layer + replicas.

2.3 DynamoDB Global Tables & Scaling

- Global Tables = multi-region, multi-active (read/write in every region) with last-writer-wins conflict resolution — for global low-latency apps + regional DR.
- Scaling is automatic (partitions split under load); On-Demand absorbs spikes instantly, Provisioned + auto-scaling tracks a target utilization. DAX adds microsecond read caching.
- Streams (24 h change log) drive replication, CDC, and event-driven pipelines via Lambda. TTL auto-expires items. Adaptive capacity + burst capacity smooth uneven access.

TIP Global Tables give both global performance and DR at once. Watch last-writer-wins semantics for concurrent cross-region writes to the same item.

2.4 Caching & Read-Scaling Strategies

- Offload reads from the primary: read replicas (relational), DAX (DynamoDB), ElastiCache (any DB query/session cache). Cache before scaling the database tier.

- RDS Proxy pools + reuses connections (critical for Lambda/serverless spikes) and improves failover time — protects the DB from connection storms.
- Write scaling is harder: shard, use queues to smooth writes, or move to Aurora/DynamoDB. Aurora offloads reads to 15 replicas; DynamoDB scales writes horizontally by partition.

WATCH Application must tolerate replica lag (eventual consistency) when reading from replicas — route read-after-write to the primary when strong consistency is required.

2.5 Backup, Restore & Point-in-Time Recovery

- RDS/Aurora automated backups enable PITR (retention 1–35 days) to any second within the window; manual snapshots persist until deleted. Restores always create a NEW instance/cluster.
- Snapshots are incremental + regional; copy cross-region/cross-account for DR; share encrypted snapshots by sharing the KMS key. Aurora backups are continuous to S3 (no perf impact).
- DynamoDB: PITR (35-day continuous) + on-demand backups (full, no perf impact). AWS Backup centralizes policy-driven backups across RDS/Aurora/DynamoDB/DocumentDB/Neptune with cross-region copy.

TIP Restoring is a new endpoint — plan DNS/connection-string cutover. Test restores regularly; a backup you've never restored is a hope, not a plan.

2.6 Disaster Recovery Strategies for Databases

- By RTO/RPO/cost: Backup & Restore (cross-region snapshots, hours), Pilot Light (cross-region read replica ready to promote), Warm Standby (scaled-down running copy), Active/Active (Global Tables / Aurora Global DB write-forwarding).
- Cross-region read replicas (RDS/Aurora) give low-RPO DR + can be promoted on failover. DynamoDB Global Tables + Aurora Global DB give the lowest RTO/RPO.

TIP Pick the DR pattern per DB tier by RTO/RPO budget: promote a cross-region replica for minutes-level RTO; Global Tables/Global DB for near-zero.

Chapter 3 — Deployment & Migration

3.1 Deployment Automation

- Provision databases as code with CloudFormation/CDK (instances, parameter/option groups, subnet groups, security groups) for repeatable, reviewable deploys across environments/regions.
- Parameter groups tune engine config; option groups add features. Subnet groups define which AZs/subnets the DB can use (place DBs in private/isolated subnets).
- Use CI/CD + change sets to preview modifications; separate config per environment (dev/test/prod) via parameters.

TIP Idempotent IaC makes DR-region rebuilds and blue/green upgrades reproducible — no click-ops drift.

3.2 Migration Strategy: Homogeneous vs Heterogeneous

- Homogeneous (same engine, e.g. on-prem SQL Server → RDS SQL Server): use native tools (backup/restore, log shipping) or DMS. Schema + data move directly.
- Heterogeneous (engine change, e.g. Oracle → Aurora PostgreSQL, SQL Server → Aurora MySQL): use SCT for schema/code conversion + DMS for data (with CDC for ongoing sync).
- Choose the migration method by downtime tolerance: snapshot/restore (some downtime), native replication/log shipping, or DMS full-load + CDC (near-zero downtime).

TIP Rule: same engine → native tools or DMS; different engine → SCT (schema/code) THEN DMS (data + CDC). Validate with DMS data validation before cutover.

3.3 AWS DMS Deep Dive

- Components: replication instance (runs the migration), source + target endpoints, and migration tasks. Supports full load, full load + CDC, and CDC-only.
- CDC (change data capture) replicates ongoing changes so the source stays live until cutover → minimal downtime. Table mappings + transformation rules select/rename/filter objects.
- DMS migrates DATA (and can do basic homogeneous schema); for heterogeneous SCHEMA conversion you still need SCT. DMS data validation compares source vs target row-by-row.
- Sizing: pick replication-instance class for throughput; watch LOB handling (limited vs full LOB mode), latency (CDCLatencySource/Target), and target load performance.

WATCH DMS is for migration/replication, not a long-term HA solution. Disable target triggers/secondary indexes during full load, then rebuild, to speed loads.

3.4 AWS Schema Conversion Tool (SCT)

- SCT converts schema + database code (stored procedures, functions, views, triggers) between heterogeneous engines and produces an Assessment Report estimating manual-conversion effort.

- Handles what can't auto-convert by flagging action items for manual rework. Pairs with DMS: SCT builds the target schema, DMS moves the data.
- SCT extraction agents + data extractors help large data-warehouse migrations (e.g. Oracle/Teradata → Redshift).

TIP Run the SCT Assessment Report first — it quantifies conversion effort and surfaces incompatibilities before you commit to a heterogeneous migration path.

3.5 Migration Patterns & Minimizing Downtime

- Near-zero-downtime pattern: SCT (schema) → DMS full load → DMS CDC keeps target in sync → validate → cut over application (DNS/connection string) → stop CDC.
- Large homogeneous SQL Server: native backup to S3 + restore on RDS (option group), or DMS. Oracle: Data Pump / DMS. Keep a rollback window post-cutover.
- For big offline datasets where the network is the bottleneck, seed with Snowball then catch up with DMS CDC.

TIP Always: assess → convert → migrate → validate → cut over → monitor. Keep the source available for rollback until the target is proven in production.

Chapter 4 — Management & Operations

4.1 Maintenance Windows & Engine Upgrades

- Maintenance window = weekly slot for AWS-applied patches/maintenance; backup window for automated backups (don't overlap heavily). Some changes apply immediately vs during the window.
- Minor version upgrades = bug/security fixes (can be auto-applied); major version upgrades = new features/possible breaking changes (manual, test first, may need parameter-group updates).
- Blue/Green Deployments create a synced green copy for upgrade/testing, then switch over in ~a minute with rollback safety — ideal for major upgrades + schema changes.

TIP Test major upgrades on a restored snapshot / blue-green green environment first. Read replicas may need upgrading in order relative to the primary.

4.2 Parameter & Option Groups, Configuration

- Parameter groups control engine settings (e.g. max_connections, work_mem, timeouts). Dynamic params apply immediately; static params require a reboot. Custom groups let you tune per workload.
- Option groups enable engine add-ons: Oracle/SQL Server TDE, SQL Server Audit, native backup/restore, Oracle APEX, MariaDB audit plugin.
- Right-size max_connections to instance memory; use RDS Proxy to avoid connection exhaustion rather than over-inflating limits.

WATCH Changing a static parameter needs a reboot to take effect — plan maintenance. Applying a new parameter group is not enough by itself for static params.

4.3 Storage Management & Scaling

- gp3 (baseline 3000 IOPS/125 MB/s, independently scalable) vs io1/io2 (provisioned IOPS for consistent high performance). Storage autoscaling grows volume before it fills.
- Aurora storage auto-grows/shrinks with data (pay for used); RDS storage can scale up but generally not down (create new instance to shrink).
- Monitor FreeStorageSpace, BurstBalance (gp2), IOPS, and throughput. Provisioned IOPS for latency-sensitive OLTP; gp3 for most general workloads.

TIP Enable storage autoscaling to prevent out-of-space outages; watch IOPS/throughput limits tied to volume type + size, not just capacity.

4.4 RDS Proxy & Connection Management

- RDS Proxy pools and multiplexes database connections — essential for serverless/Lambda where thousands of short-lived connections can exhaust the DB.

- Benefits: reduced failover time (holds connections, reconnects to new primary), IAM authentication, and Secrets Manager-managed credentials (no creds in code).
- Supports RDS/Aurora MySQL & PostgreSQL. Deploy in the same VPC/subnets; connection pinning can reduce reuse (watch session-state statements).

TIP Lambda + RDS → almost always add RDS Proxy to prevent connection storms and speed up failover.

4.5 Operational Automation

- Automate with EventBridge (react to RDS events), Lambda (custom actions), and Systems Manager (run commands, maintenance automation, Parameter Store for config).
- RDS event subscriptions (via SNS) notify on failovers, low storage, backups, and configuration changes. AWS Backup applies org-wide backup policies.
- Tag resources for cost allocation + automation targeting; use Config rules to enforce encryption/backup/Multi-AZ compliance across DB fleets.

TIP Event-driven ops: RDS event → EventBridge rule → Lambda/SNS for auto-remediation and alerting (e.g. auto-notify + scale storage on low-space events).

Chapter 5 — Monitoring & Troubleshooting

5.1 CloudWatch Metrics for Databases

- Key RDS/Aurora metrics: CPUUtilization, DatabaseConnections, FreeableMemory, FreeStorageSpace, Read/WriteIOPS, Read/WriteLatency, ReplicaLag, and (Aurora) AuroraReplicaLag, VolumeBytesUsed.
- DynamoDB metrics: ConsumedRead/WriteCapacityUnits, ThrottledRequests, ReadThrottleEvents, SuccessfulRequestLatency, SystemErrors. Set alarms on throttling + latency.
- Alarms → SNS/EventBridge for alerting + auto-remediation. Dashboards consolidate fleet health.

TIP ReplicaLag spike = replication bottleneck; FreeStorageSpace trending down = autoscale/clean up; DatabaseConnections near max = add RDS Proxy.

5.2 Performance Insights & Enhanced Monitoring

- Performance Insights = database-load dashboard measuring Average Active Sessions (AAS) vs vCPU line, with top SQL, top waits, and top hosts/users — the primary tuning tool. Retention up to 2 years (paid) / 7 days free.
- Enhanced Monitoring = OS-level metrics (per-second, via an agent): CPU, memory, file system, per-process — deeper than CloudWatch's hypervisor-level metrics.
- Use Performance Insights to find the query/wait event driving load, then Enhanced Monitoring to confirm OS-level resource pressure.

TIP AAS above the vCPU line = the DB is a bottleneck; drill into top SQL + top wait events to target tuning (I/O, lock, CPU, or commit waits).

5.3 RDS / Aurora Troubleshooting

- High CPU → find heavy queries (Performance Insights), add indexes, tune parameters, or scale up/out (read replicas). High connections → RDS Proxy / connection pooling.
- Storage full → enable autoscaling, purge/archive, check binlog/WAL retention. Replica lag → long transactions on primary, under-sized replica, or heavy write bursts.
- Failover events → check the RDS events + application retry logic + DNS TTL; ensure the app reconnects to the cluster endpoint, not a cached instance IP.

WATCH Long-running transactions and uncommitted work inflate WAL/binlog and replica lag — watch for idle-in-transaction sessions.

5.4 DynamoDB Troubleshooting

- Throttling (ProvisionedThroughputExceeded / ThrottledRequests) → hot partition key, insufficient capacity, or a GSI under-provisioned relative to the base table.
- Fixes: redesign partition key for higher cardinality, switch to On-Demand, raise provisioned capacity/auto-scaling targets, add DAX for read hotspots, or spread writes.

- Watch GSI capacity separately (a throttled GSI can throttle base-table writes). Use exponential backoff + jitter in the SDK for retries.

TIP Most DynamoDB pain traces to key design. High-cardinality, evenly-accessed partition keys are the cure for hot partitions + throttling.

5.5 Logging & Auditing

- Publish DB engine logs (error, slow query, general, audit) to CloudWatch Logs for retention + analysis. Aurora/RDS support exporting logs; use Logs Insights to query.
- Auditing: MariaDB/MySQL audit plugin, SQL Server Audit (SQLSERVER_AUDIT option), Oracle unified auditing, PostgreSQL pgAudit — configured via option/parameter groups.
- CloudTrail records control-plane API calls (create/modify/delete DB); Database Activity Streams (Aurora) give a near-real-time, protected stream of data-plane activity for compliance.

TIP Control-plane audit (who changed the instance) = CloudTrail; data-plane audit (who queried the data) = engine audit logs / Database Activity Streams.

Chapter 6 — Database Security

6.1 Encryption at Rest & In Transit

- At rest: KMS encrypts storage, automated backups, snapshots, and read replicas. Must enable at creation — to encrypt an existing unencrypted DB, snapshot → copy (encrypt) → restore.
- In transit: enforce TLS/SSL (download the RDS CA bundle; PostgreSQL `rds.force_ssl`, SQL Server force encryption, MySQL require SSL). Aurora supports TLS by default.
- TDE (Transparent Data Encryption) for Oracle/SQL Server via option groups (engine-level encryption in addition to KMS volume encryption).

WATCH You cannot directly toggle encryption on an existing RDS instance — the snapshot-copy-with-encryption-then-restore path is the exam-favorite answer.

6.2 IAM Authentication & Access Control

- IAM database authentication (MySQL/PostgreSQL/Aurora) uses short-lived tokens (15 min) instead of passwords — no stored DB passwords, centralized via IAM. Good for apps/roles; watch connection-rate limits.
- Control-plane access (create/modify/delete DB) is governed by IAM policies; data-plane access (SQL privileges) is managed inside the database engine (GRANT/REVOKE).
- DynamoDB uses IAM natively for data access (fine-grained: restrict to specific items/attributes via condition keys like `dynamodb:LeadingKeys`).

TIP IAM auth removes password management for RDS/Aurora; combine with RDS Proxy + Secrets Manager for the cleanest credential story.

6.3 Network Security

- Place databases in private/isolated subnets (no direct internet route). DB subnet group defines the AZs/subnets. Security groups (stateful) restrict which app-tier SGs can reach the DB port.
- Reference the app-tier security group as the source in the DB SG (SG-to-SG) rather than IP ranges. Use VPC endpoints for private access to APIs (Secrets Manager, KMS) without internet.
- For hybrid/on-prem access use VPN/Direct Connect; never expose a DB publicly. RDS 'publicly accessible = No' keeps it internal.

WATCH A publicly accessible RDS instance + open security group is a top misconfiguration — keep DBs private and SG-scoped to the app tier.

6.4 Secrets Management & Rotation

- AWS Secrets Manager stores DB credentials and rotates them automatically (built-in rotation Lambdas for RDS/Aurora/DocumentDB/Redshift). RDS Proxy integrates directly with Secrets Manager.
- SSM Parameter Store (SecureString) is a cheaper option for config/secrets without native rotation.

- Never hard-code credentials; grant the app role read access to the specific secret; rotate on a schedule + on compromise.

TIP Secrets Manager when you need automatic rotation + integration (RDS Proxy); Parameter Store for simple, low-cost config/secrets.

6.5 Auditing, Compliance & Governance

- Enable engine auditing (SQL Server Audit, pgAudit, MySQL/MariaDB audit plugin, Oracle auditing) via option/parameter groups; ship logs to CloudWatch/S3 for retention.
- AWS Config rules enforce DB compliance (encryption enabled, Multi-AZ, backups, not public); CloudTrail provides the API audit trail; Database Activity Streams for protected data-plane monitoring.
- Compliance: RDS supports HIPAA/PCI/SOC/FedRAMP eligibility; use AWS Artifact for reports. Least-privilege IAM + KMS key policies scope access.

TIP Layered DB security = private network + SG scoping + KMS at rest + TLS in transit + IAM/Secrets Manager auth + engine audit logging + Config compliance.

Chapter 7 — Certification Status, Reference & Interview Prep

7.1 DBS-C01 Status & Domain Review

- **IMPORTANT:** The AWS Certified Database – Specialty (DBS-C01) was RETIRED on April 30, 2024. New candidates can no longer sit it; existing holders stay certified for 3 years from their pass date.
- No direct replacement was announced. The closest active credential is AWS Certified Data Engineer – Associate (DEA-C01), but it targets data pipelines/engineering — not deep DBA administration.
- The DBS-C01 domains still map perfectly to Cloud DBA interviews: (1) Workload-Specific DB Design 26%, (2) Deployment & Migration 20%, (3) Management & Operations 18%, (4) Monitoring & Troubleshooting 18%, (5) Database Security 18%.

TIP Use this material as an interview/knowledge reference. For an active cert on a similar track, look at DEA-C01 (data engineering) or SAA-C03/SAP-C02 (architecture).

7.2 Service Selection Decision Guide

- 'Complex joins / OLTP / existing SQL' → RDS or Aurora. 'High throughput + cloud-native failover + 15 readers' → Aurora. 'Oracle/SQL Server managed' → RDS (or RDS Custom for OS access).
- 'Known access patterns + massive scale + single-digit ms' → DynamoDB. 'Microsecond reads on DynamoDB' → DAX. 'Cache / sessions / leaderboards' → ElastiCache Redis. 'Durable in-memory primary' → MemoryDB.
- 'Graph' → Neptune. 'Document/JSON (Mongo)' → DocumentDB. 'Cassandra' → Keyspaces. 'Time-series' → Timestream. 'Immutable ledger' → QLDB. 'OLAP/warehouse' → Redshift. 'Search' → OpenSearch.
- 'Migrate same engine' → native tools/DMS. 'Change engine' → SCT + DMS. 'Cross-region DR' → cross-region read replica / Aurora Global DB / DynamoDB Global Tables.

7.3 Key Limits & Numbers Reference

- Aurora: 6 storage copies / 3 AZs, up to 128 TiB, 15 replicas, ~30 s failover, Global DB <1 s replication. RDS Multi-AZ failover ~60–120 s (classic), ~35 s (Multi-AZ DB cluster).
- DynamoDB: 400 KB item, 1 RCU = 4 KB strong read/s, 1 WCU = 1 KB write/s, LSI 10 GB item-collection cap, Streams 24 h retention, PITR 35 days.
- Backups: RDS/Aurora automated retention 1–35 days (PITR); manual snapshots kept until deleted. IAM auth token lifetime 15 min.

TIP These numbers show up constantly in scenario questions and interviews — memorize the Aurora/DynamoDB/backup figures cold.

7.4 Interview Strategy & Career Path

- Answer database scenarios by stating: workload type (OLTP/OLAP/NoSQL), consistency/latency needs, scale, HA/DR requirement (RTO/RPO), and cost — then justify the engine choice against those.
- Frequent interview themes: Multi-AZ vs read replica, encrypt an existing RDS (snapshot-copy path), heterogeneous migration (SCT+DMS+CDC), throttling/hot-partition fixes, Performance Insights tuning, RDS Proxy for serverless.

- For a Cloud DBA path: pair this DB depth with SAA-C03 (architecture) and/or DEA-C01 (data engineering); hands-on with DMS, Performance Insights, and Aurora Global DB cements the concepts.

TIP Senior DBA interviews reward trade-off reasoning + operational war stories (migrations, failovers, tuning) over memorized definitions — lead with the 'why'.